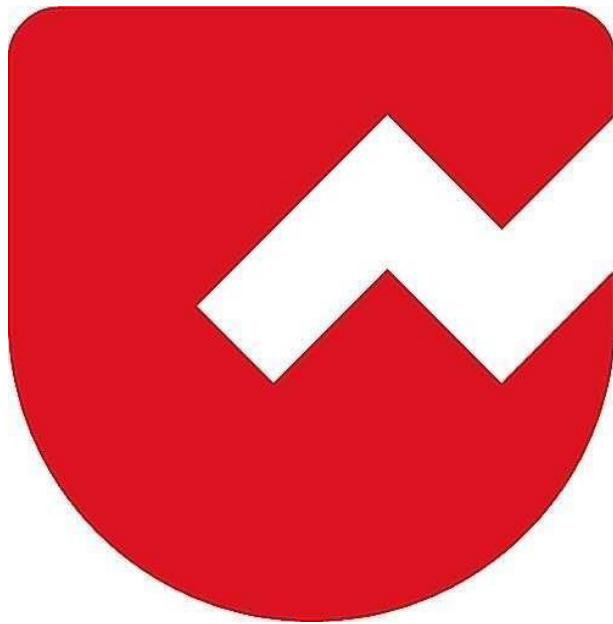




Zoho Schools for Graduate Studies



Notes

WHAT IS JAVA ?

Java is a

- Simple,
- Powerful,
- Platform Independent,
- Purely Object Oriented,
- Multi – Threaded,
- High level Programming language.

Explanation for these Features :

Simple – It retains the Syntax of c and C++ syntax.

Powerful - Java is powerful because it is **platform-independent, object-oriented, secure, robust, and supported by rich libraries with strong community support.**

Why java becomes platform independent ?

Java follows WORA (Write Once, Run Anywhere) architecture.

Byte codes made java platform independent because it can run on any system with a JVM.

In java , the source code is compiled by the Java compiler into intermediate bytecode, the JVM translates the byte code to platform specific code.

Note : any language can achieve platform independence if it uses an intermediate representation + a virtual machine (just like Java).

What Qualifies the language to be Object – Oriented ?

The language qualifies to be Object Oriented if and only if it satisfies the three object oriented principles :

- Encapsulation
- Inheritance
- Polymorphism

Multi-Threaded :

Java have the important feature of multi threading where we can do multiple tasks concurrently,with in the same program.

High Level Programming Language :

Instructions Given to the machines in user – friendly / user – understandable language

Note : Low level Programming languages , where the instructions where given to the machines in machine understandable form (like 0's and 1's)

Additional Info on other languages :

1) why Java Script is Object based language ?

Java Script supports Encapsulation have only classes and objects , but does not support inheritance. So java script is object bases language.

2) Why C++ is partially object Oriented ?

C++ is partially object-oriented because we can write code with classes or without classes. Java is fully object-oriented because everything must be inside a class.

History Of Java :

- **1991** → James Gosling, Patrick Naughton, and Mike Sheridan at Sun Microsystems started a project called the Green Project.
- **Objective of Green Project** → To create a language for small electronic devices (like remote controls) that could be platform-independent.
- The first language they made was called Oak (named after an oak tree outside Gosling's office).
- Later, **Oak** had to be renamed because another company already had a product with that name.
- They chose the name **Java** because the team drank a lot of Java coffee, and they wanted a short, unique, and catchy name

Java Editions :

1. Java Platform, Standard Edition (Java SE)
2. Java Platform, Enterprise Edition (Java EE)
3. Java Platform, Micro Edition (Java ME)

Features of java :

Java is Secure :

- Java uses its own **JRE**, part of **JVM (Sandbox)** for execution.
- Predecessor languages used the **runtime environment of the operating system**.
- **Class loader** loads classes into JVM, and **Bytecode verifier** checks for malicious code.

Java is Robust

- Robust means “**Strong**”.
- Memory management mistakes and runtime errors cause program failures.
- Automatic memory management prevents explicit harm to memory.
- Exception handling addresses mishandled runtime error

Interpreted and High Performance

- **Interpreter** converts bytecode into machine-readable binary code.
- Java uses **JIT (Just-In-Time compiler)** for very high performance.
- Interpreters are relatively slower, but **JIT compiler improves processing speed**.

Portable

- Java programs can run across different platforms without modification.
- Platform independence + standard libraries make Java portable.

Architecture Neutral

- Bytecode is designed to be independent of processor architecture.

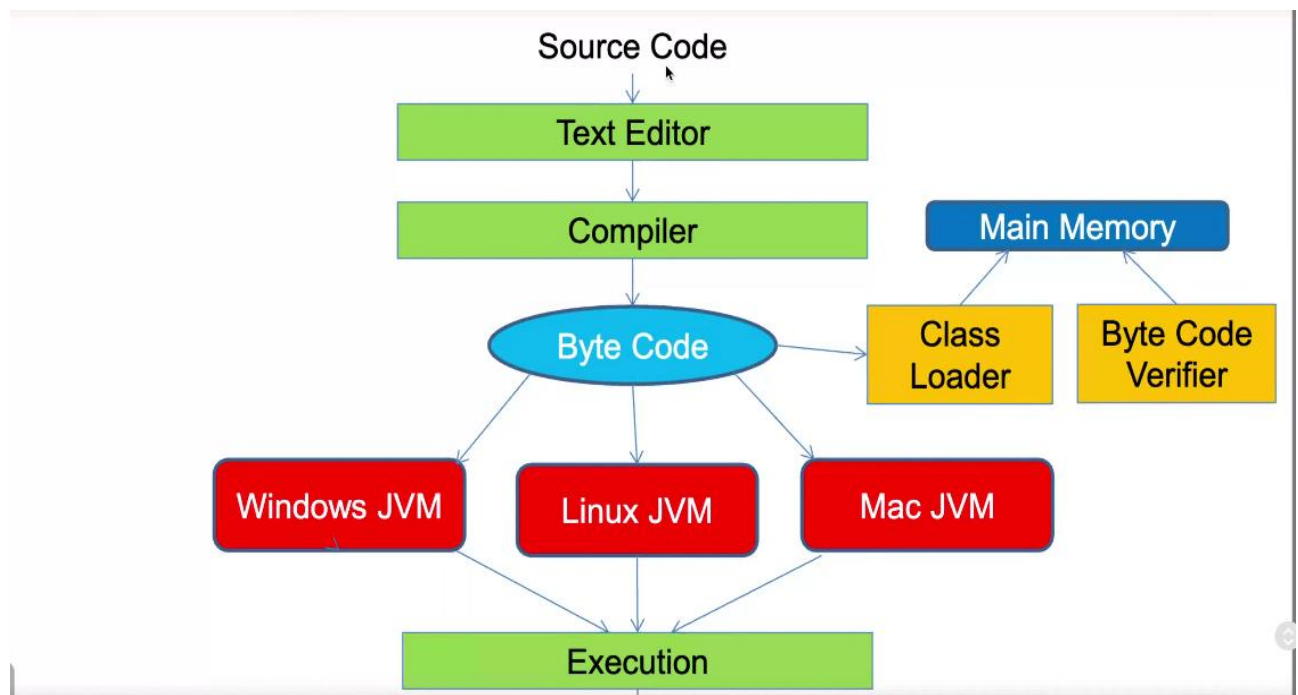
Dynamic & Extensible

- Classes can be loaded dynamically at runtime.
- Java supports integration with other languages (e.g., C/C++ using JNI).

Uses of Java

- **Desktop Applications** → e.g., calculators, media players, IDEs
- **Web Applications** → e.g., e-commerce sites, online portals
- **Mobile Applications** → especially Android apps (since Android uses Java).
- **Networking Applications** → e.g., chat apps, email servers, online games.
- **Enterprise Applications** → e.g., banking software, ERP, large-scale business systems.
- **Scientific Applications** → e.g., simulations, research tools.
- **Game Development** → both desktop and mobile games.

Working of Java Program Execution



- Java program is written as **source code** using a text editor.
- The source code (.java file) is compiled by the **Java Compiler** into **Byte Code** (.class file).

- **Byte Code** is platform-independent and can run on any operating system with JVM.
- The **Class Loader** loads the byte code into **main memory**.
- The **Byte Code Verifier** checks the byte code for validity and security.
- The **Java Virtual Machine (JVM)**, specific to each OS (Windows JVM, Linux JVM, Mac JVM), converts the byte code into machine code.
- The machine code is executed by the CPU, producing the program's output.